“From farm to factory : A technical exploration of dairy processing at Bharat dairy”

At

Bharat Horns

Summer Internship Project

Submitted:

In the partial fulfilment of the Degree of

Master of Business Administration

Semester - III

Submitted by:

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Ganpat University | V.M. Patel College of Management Studies

CERTIFICATE OF THE COMPANY

Bharat Horns
Dairy Products

Certificate of completion

This is to certify that **Ms. Sakshiben Patel Kamlesh Kumar, a** student of the **MBA Department from Ganpat University-V.M Patel Institute of Management**, has successfully completed the Digital Marketing Internship program offered by **Mehsana Dairy & food products Ltd**. This program consisted of an offline training module and an internship, and was conducted from **19th May,2025 to 18th July,2025.**

During this internship, he demonstrated a keen interest in digital marketing concept and exhibited excellent knowledge of various digital marketing strategies. He completed a summer internship project on the topic: "**From farm to factory: A technical exploration of dairy processing at Bharat Dairy**" during this internship under the guidance of Mr. Gaurav Mishra, Head of Digital Marketing.

Throughout the program, he consistently displayed creativity, and a solid understanding of industry best practices. His dedication, enthusiasm, and ability to work effectively in a team environment were commendable.

By successfully completing the internship, he has shown a commitment to learning and professional growth in the field of digital marketing. We believe his skills and accomplishments during this program will contribute significantly to his future career prospects.

We hereby present this certificate as recognition of **Ms. Sakshiben Patel Kamlesh Kumar** successful completion of the Digital Marketing Internship Program.

Date:18th July,2025

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CERTIFICATE OF THE GUIDE

This is to certify that the contents of this report entitled **"From farm to factory : A technical exploration of dairy processing at Bharat dairy"** by **Patel Sakshi K. [24044311137]** submitted to Ganpat University – V.M Patel Institute of Management for the Award of Master of Business Administration (MBA Sem III) is original research work carried out by him/her/them under my supervision.

This report has not been submitted either partly or fully to any other University or Institute for award of any degree or diploma.

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CANDIDATE'S STATEMENT

I hereby declare that the work incorporate in this report entitled "From farm to factory : A technical exploration of dairy processing at Bharat dairy" in partial fulfillment of the requirements for the award of Master of Business Administration (Semester-III) is the outcome of original study undertaken by me and it has not been submitted earlier to any other University or Institution for the award of any Degree or Diploma.

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PREFACE

The dairy industry in India has seen a remarkable transformation from a traditional farm – based system to a modern industrial setup. This Project titled " From Farm to Factory: A Technical Exploration of Dairy Processing at Bharat Dairy." Explores the journey of milk from the farm to the final packaged product. The project emphasizes the technological, operational, and logistical aspects involved in dairy processing at Bharat Dairy, a division of Bharat Horns, Mehsana. This research also explores market trends, government initiatives, and challenges facing the industry.

ACKNOWLEDGEMENT

I take this opportunity to express my deep sense of gratitude to all those who supported and guided me throughout the successful completion of this summer internship project titled *"From Farm to Factory: A Technical Exploration of Dairy Processing at Bharat Dairy."*

First and foremost, I would like to express my heartfelt thanks to **Mr. Meet Shah**, my project guide, for his invaluable guidance, constant encouragement, and thoughtful feedback. His support helped me stay focused and complete this project with clarity and precision.

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Patel Sakshiben Kamleshkumar
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EXECUTIVE SUMMARY

Chapter 1: Introduction

Chapter 1 provides a comprehensive overview of the dairy industry, focusing on the complete farm-to-factory journey of milk at Bharat Dairy. The chapter highlights the technical steps involved in milk procurement, transportation, quality testing, pasteurization, homogenization, packaging, and cold-chain distribution. It discusses global and Indian dairy scenarios, emphasizing growth trends, technological advancements, and sustainability practices. Analytical tools such as SWOT, PESTEL, and Porter's Five Forces are used to understand the competitive environment. The chapter concludes by identifying critical success factors like quality control, operational efficiency, technological adoption, and supply-chain strength, which are essential for dairy industry competitiveness.

Chapter 2: Introduction of the Company

Chapter 2 details the background, evolution, and organizational structure of Bharat Horns (Mehsana Dairy & Food Products Ltd.). It explains how the company grew from a small milk collection initiative to a large-scale dairy processing unit handling nearly 400,000 liters per day. The chapter also presents the company's pillars—Purity, Innovation, Diversification, Sustainability, Customer Trust, and Expansion Strategy—showing how these core values guide operations. A detailed company history highlights expansion in product lines, technological upgrades, and sustainability initiatives. The SWOT analysis of the company summarizes internal strengths and weaknesses, as well as external opportunities and challenges, positioning Bharat Dairy as a fast-growing and quality-driven dairy brand in Gujarat.

Chapter 3: Literature Review

Chapter 3 synthesizes previous research related to dairy processing, focusing on five major variables: Product Quality, Operational Efficiency, Customer Satisfaction, Quality Control Practices, and Factory Technical Factors. The literature review reveals significant gaps in studies conducted on medium-scale private dairies, particularly concerning technical processes, sustainability practices, and operational integration. It provides theoretical foundations through global and

national studies, showing how these variables interact and influence dairy processing outcomes. Hypothesis development is completed based on the theoretical framework, and a conceptual research model is presented to guide the empirical investigation.

Chapter 4: Research Methodology

Chapter 4 outlines the methodological framework used in the study. It includes the problem statement, research objectives, scope of the study, and detailed research design. The study follows an analytical and exploratory design based on structured questionnaires administered to 50 respondents across various departments of Bharat Dairy. The chapter presents the variables, research constructs, sampling design, data collection methods, and analysis tools including SPSS for descriptive statistics, reliability testing, and regression analysis. The questionnaire layout, including demographic details and variable-wise items based on validated literature, ensures reliability and accuracy of measurement.

Chapter 5: Data Analysis & Interpretation

Chapter 5 presents detailed statistical analysis of the collected data. Descriptive statistics summarize demographic profiles, while item analysis shows strong positive responses toward product quality, operational efficiency, customer satisfaction, and quality control practices. Cronbach's Alpha values for all constructs exceed recommended standards, confirming high reliability of the instrument. Regression and mediation analyses reveal significant relationships among the variables. Product Quality, Operational Efficiency, and Customer Satisfaction all significantly influence Quality Control Practices, which in turn strongly impact Factory Technical Factors. Mediation models further confirm that Quality Control Practices play a central and mediating role in enhancing factory technical performance.

Chapter 6: Conclusion, Findings & Recommendations

Chapter 6 summarizes the results of the study, concluding that quality control practices serve as the most critical link in ensuring superior technical performance within the factory. Key findings indicate that high product quality, efficient operations, and strong customer satisfaction significantly enhance quality control

mechanisms. Recommendations include adopting advanced digital monitoring, expanding automation, strengthening cold-chain management, training workers on hygiene and handling practices, and implementing sustainability-driven initiatives. The chapter concludes that Bharat Dairy is well positioned for continued growth by leveraging technology, quality enhancement, and operational excellence.

CONTENT

Table No.	Particular	Page No.
	CERTIFICATE OF THE COMPANY	**II**
	CERTIFICATE OF THE GUIDE	**III**
	CANDIDATE'S STATEMENT	**IV**
	PREFACE	**V**
	ACKNOWLEDGEMENT	**VI**
	EXECUTIVE SUMMARY	**VII**
1	**INTRODUCTION**	**12**
1.1	OVERVIEW OF TOPIC	13
1.2	GLOBAL SCENARIO	15
1.3	INDIAN SCENARIO	16
1.4	GROWTH OF THE INDUSTRY	17
1.5	CRITICAL SUCCESS FACTORS OF INDUSTRY	18
1.6	INDUSTRY SWOT ANALYSIS	19
1.7	PESTEL ANALYSIS	21
1.8	MICHAEL PORTER'S FIVE FORCE ANALYSIS	22
2	**INTRODUCTION OF THE COMPANY**	**25**
2.1	OVERVIEW OF THE COMPANY	26
2.2	COMPANY HISTORY	27
2.3	SIX PILLERS OF COMPANY	28
2.4	ORGANIZATION STRUCTURE	29
2.5	SWOT ANALYSIS	30
3	**LITERATURE REVIEW**	**32**
3.1	RESEARCH GAP	33
3.2	STUDY VARIABLE GLOBAL CONTEXT	33
3.3	HYPOTHESIS DEVELOPMENT	36
3.4	RESERCH MODEL	38
4	**RESERCH METHODOLOGY**	**38**
4.1	PROBLEM STATEMENT	39

4.2	RESEARCH OBJECTIVE	39
4.3	SCOPE OF THE STUDY	40
4.4	RESEARCH DESIGN	41
4.5	RESEARCH INSTRUMENT	41
4.6	SAMPLING DESIGN	44
4.7	SOURCE OF DATA	45
5	**DATA ANALYSIS**	**47**
5.1	INTRODUCTION	48
5.2	ITEM ANALYSIS	49
5.3	RELIABILTY TEST	54
5.4	REGRESSION ANALYSIS	55
6	**CONCLUSION, FINDINGS & RECOMMENDATION**	**64**
6.1	CONCLUSION	65
6.2	FINDINGS	65
6.3	RECOMMENDATION	66
	REFERENCES	**68**
	APPENDIX	**71**

CHAPTER 1

INTRODUCTION

1.1 Overview of Topic

The study titled *Technical Exploration of Dairy Processing at Bhart Dairy* provides a comprehensive understanding of how raw milk travels through a structured farm-to-factory system before becoming safe and high-quality dairy products for consumers. It begins at the farm level, where proper milking techniques, udder hygiene, and immediate cooling through Bulk Milk Coolers ensure that the milk retains freshness and low microbial levels.

From there, Bhart Dairy's procurement network collects and transports the milk in insulated tankers, maintaining the cold chain and preventing spoilage. Upon arrival at the plant, the raw milk undergoes a series of rigorous quality checks—including fat, SNF, pH, temperature, adulteration, and microbial tests—to ensure it meets strict safety standards.

The milk then moves through key processing stages such as filtration, clarification, standardization, pasteurization through HTST technology, and homogenization, all of which help maintain product uniformity, destroy harmful microorganisms, and enhance texture.

Bhart Dairy also produces multiple value-added products such as curd, paneer, lassi, ghee, and butter using controlled fermentation systems, cream separators, coagulation tanks, and ghee boilers. After processing, fully automated packaging systems using food-grade multilayer films, batch coding, and aseptic techniques ensure tight hygiene and traceability. The packaged dairy products are stored in temperature-controlled cold rooms and distributed through insulated vehicles to retain freshness until they reach retailers and consumers.

The study also highlights Bhart Dairy's focus on sustainability through wastewater treatment, energy-efficient equipment, digital monitoring systems, and waste-reduction technologies. Overall, this detailed exploration shows how Bhart Dairy integrates modern machinery, quality control protocols, regulatory compliance, and efficient logistics to transform raw milk into safe, standardized, and high-quality dairy products ready for the market.

Packaging forms another crucial stage in dairy processing where Bhart Dairy uses automatic pouch-packing machines, aseptic filling systems, and food-grade multilayer

films to protect milk and value-added products from contamination. Batch coding and labeling provide traceability and compliance with FSSAI and ISO 22000 standards. The dairy emphasizes strict sanitation practices through Clean-in-Place (CIP) systems that automatically clean pipelines, storage tanks, and processing equipment with hot water and chemicals. This ensures hygiene, prevents cross-contamination, and reduces manual labor. The entire packaging and sanitation system reflects Bhart Dairy's adherence to Good Manufacturing Practices (GMP) and HACCP principles.

After packaging, the dairy products are transferred to large cold storage rooms maintained at specific temperatures depending on the product type. Refrigerated chambers for milk, fermentation rooms for curd, and deep-freeze units for butter and ghee ensure each product maintains its stability until dispatch. For distribution, Bhart Dairy uses insulated refrigerated vehicles that maintain required temperatures during transportation to retail points, supermarkets, and distribution hubs. This logistics network ensures that the final product delivered to consumers is fresh, safe, and of high quality.

A notable part of Bhart Dairy's approach is its commitment to sustainability and environmental responsibility. The dairy operates effluent treatment plants to treat wastewater before discharge and uses energy-efficient boilers, heat recovery systems, and solar energy solutions to reduce resource consumption. Waste such as whey is recycled or repurposed, reducing environmental impact. The plant also adopts digital technologies such as SCADA systems for monitoring operations, machine performance, and temperature control. These innovations demonstrate Bhart Dairy's dedication to modern, eco-friendly processing systems.

In summary, the technical exploration of Bhart Dairy reveals an integrated and scientifically controlled system that ensures the safe journey of milk from farms to factory shelves. The study highlights how Bhart Dairy combines modern technology, strict quality checks, advanced machinery, traceability, hygienic handling, and sustainability practices to maintain high standards in dairy processing. This four–five-page overview underscores that the quality of Bhart Dairy products is not accidental—it is the result of well-coordinated processes, skilled management, and constant technological innovation across the entire dairy supply chain.

1.2 GLOBAL SCENARIO

The global dairy industry is a highly dynamic and technologically advanced sector that plays a critical role in food security, nutrition, and rural livelihoods. Worldwide, milk production has been steadily increasing due to rising population, growing urbanization, and higher consumer awareness of dairy's nutritional benefits. Countries like the United States, India, China, and members of the European Union are the leading producers of milk, collectively contributing to the majority of global dairy output. Modern dairy processing across the world follows similar principles of farm-to-factory integration, emphasizing hygiene, quality control, and value addition, though the level of technology and automation varies regionally. In developed countries, dairy processing is dominated by highly automated plants that use state-of-the-art pasteurization, homogenization, fermentation, and packaging systems, often employing robotics and Industry 4.0 technologies for process monitoring, traceability, and efficiency. Value-added dairy products such as flavored milk, yogurt, cheese, whey protein, and milk powders are significant contributors to industry revenue, driven by consumer demand for convenience and nutrition.

In developing countries, including India, the global scenario is marked by a transition from traditional milk handling to organized industrial processing. Milk procurement networks are being strengthened through cooperative societies, dairy federations, and private enterprises, enabling smallholder farmers to access modern chilling, storage, and quality testing technologies. The global dairy market also reflects a strong focus on sustainability, with many countries implementing eco-friendly practices such as wastewater treatment, renewable energy use, packaging innovation, and waste minimization. Additionally, international regulatory frameworks and quality standards—such as ISO 22000, HACCP, and Codex Alimentarius—set benchmarks for safety, traceability, and hygiene, which influence operations even at the local dairy level.

From a technological perspective, the global dairy industry is embracing innovations such as high-efficiency pasteurizers, microfiltration, ultrafiltration, automated fermentation systems, aseptic packaging, and digital monitoring tools. These advancements not only improve product safety and shelf life but also enhance productivity and reduce post-harvest losses. Consumer trends worldwide are further

driving diversification into functional dairy products, organic milk, lactose-free options, and fortified beverages, creating opportunities for both large-scale and small-scale dairy enterprises.

In this global context, Bhart Dairy aligns itself with international best practices by adopting modern processing technology, cold chain management, and strict quality control measures, while also addressing regional supply chain challenges. Its operations reflect a blend of global technological standards and local dairy development practices, positioning it competitively in both domestic and international perspectives of milk processing. The global scenario emphasizes that successful dairy processing requires not only technical capability but also an integrated approach to production, quality assurance, sustainability, and consumer-focused innovation.

1.3 Indian Scenario

India holds a unique position in the global dairy industry as the world's largest milk producer, contributing more than 22% of the global output. The dairy sector in India is a critical component of rural livelihoods, providing employment to millions of smallholder farmers, particularly women, and supporting economic growth in both rural and semi-urban areas. Unlike many developed countries, India's dairy production is dominated by small-scale farmers who typically own one or two milch animals, making the organization of the supply chain both a challenge and a priority. Over the past few decades, the sector has undergone significant transformation due to the cooperative movement pioneered by Amul and state-level milk federations, as well as the entry of private players such as Bhart Dairy, Mother Dairy, and Heritage Foods.

The Indian dairy processing industry has shifted from traditional milk handling to organized processing units that emphasize quality, hygiene, and value addition. Modern Indian dairies employ technologies such as high-capacity pasteurizers, homogenizers, automated cream separators, fermentation tanks, and aseptic packaging systems to ensure that milk and its derivatives meet domestic and international safety standards. Cold chain management has emerged as a critical factor in reducing post-harvest losses, with insulated tankers, chilling centers, and temperature-controlled storage becoming standard practices in urban and semi-urban supply networks. Value-added products, including curd, paneer, flavored milk, ghee, butter, and dairy-based beverages, have experienced growing demand due to

urbanization, rising disposable incomes, and shifting consumer preferences toward convenience and nutrition.

Regulatory frameworks play an essential role in ensuring product safety and standardization. Organizations such as FSSAI (Food Safety and Standards Authority of India), BIS (Bureau of Indian Standards), and ISO-accredited systems provide guidelines for quality testing, traceability, packaging, and labeling, which Indian dairies like Bhart Dairy strictly follow. Additionally, there is increasing emphasis on sustainability practices, such as energy-efficient equipment, wastewater treatment, and by-product utilization (e.g., whey conversion), reflecting global best practices adapted to local conditions.

In recent years, technological innovations, digitization, and cold chain expansion have transformed the Indian dairy sector into a more organized, efficient, and quality-driven industry. However, challenges such as fragmented production, seasonal supply fluctuations, and limited farmer awareness about advanced practices remain. Despite these challenges, India's dairy sector continues to grow steadily, positioning it as a global leader while ensuring nutrition security, rural development, and economic empowerment. Bhart Dairy exemplifies this transformation by integrating modern processing technologies, rigorous quality control, and robust supply chain management, reflecting the broader trends and potential of the Indian dairy scenario.

1.4 Growth of the Industry

The dairy industry, both globally and in India, has witnessed significant growth over the past few decades due to increasing demand for milk and milk-based products driven by population growth, urbanization, rising incomes, and greater awareness of nutrition and health benefits. Globally, milk production has increased steadily, with countries such as India, the United States, China, and members of the European Union leading in output. Technological advancements in dairy farming, processing, and cold chain logistics have enabled large-scale production and reduced post-harvest losses, contributing to the steady expansion of the sector. The introduction of value-added dairy products, such as flavored milk, yogurt, cheese, butter, ghee, and fortified beverages, has further propelled growth by catering to changing consumer preferences for convenience, taste, and health.

In India, the dairy industry has experienced remarkable growth, transforming from a largely unorganized sector dominated by small-scale producers to a more structured and technologically advanced industry. The success of the cooperative movement, particularly Amul's model, along with the entry of private players like Bhart Dairy, has created a reliable market for smallholder farmers, ensuring consistent income and incentivizing improvements in milk quality. Technological interventions such as bulk milk coolers, automated quality testing, high-capacity pasteurizers, and cold chain-enabled transport have enabled the industry to scale efficiently. The government's support through initiatives like the National Dairy Plan, Operation Flood, and subsidies for dairy infrastructure has also accelerated growth by modernizing dairy farms, expanding processing capacities, and improving supply chain management.

The industry's growth is further bolstered by increasing consumer demand for processed, hygienic, and convenient dairy products, urbanization, and rising health consciousness. Exports of dairy products, particularly milk powders, whey products, ghee, and cheese, are also contributing to revenue growth, positioning India as a key player in the global dairy market. Sustainability and innovation—such as energy-efficient processing, waste management, and product diversification—are emerging as important drivers that ensure the industry continues to grow responsibly. Overall, the dairy sector's growth reflects a successful integration of technology, farmer participation, policy support, and market-driven innovation, making it one of the fastest-growing segments in the agribusiness and food processing industries.

1.5 Critical Success Factors of Industry

The success and sustainability of the dairy industry, both globally and in India, depend on several interrelated factors that ensure quality, efficiency, and competitiveness. One of the most important factors is **quality of raw milk**, which begins at the farm level. Proper animal husbandry, hygiene during milking, timely chilling, and prevention of adulteration are essential to ensure that milk entering the processing chain meets safety and compositional standards. Without high-quality raw material, even advanced processing technology cannot deliver safe and consistent products.

Another critical factor is **technological capability** in processing and packaging. Modern dairies rely on pasteurization, homogenization, separation, fermentation, and automated

packaging systems to maintain microbiological safety, improve shelf life, and create value-added products. The ability to adopt and integrate advanced machinery, digital monitoring systems, and automation determines a dairy's operational efficiency and product consistency.

Efficient supply chain management and cold chain integration also play a pivotal role in success. Timely procurement, proper transportation, temperature-controlled storage, and organized distribution networks reduce post-harvest losses and ensure that products reach consumers in fresh condition. In India, where milk production is highly fragmented among smallholder farmers, establishing strong procurement networks and maintaining a robust cold chain are key differentiators.

Regulatory compliance and food safety standards constitute another critical success factor. Adhering to frameworks such as FSSAI, ISO 22000, HACCP, and BIS ensures consumer safety, enhances brand credibility, and facilitates entry into domestic and international markets. Dairies that rigorously implement quality control protocols, batch traceability, and periodic audits are more likely to gain consumer trust.

Innovation and product diversification are increasingly vital for staying competitive. The ability to develop value-added products like flavored milk, curd, paneer, ghee, yogurt, and fortified beverages, along with the adoption of eco-friendly and energy-efficient practices, allows dairies to cater to changing consumer preferences and improve profitability.

Finally, **farmer engagement and socio-economic sustainability** are central to long-term success. Providing fair pricing, technical support, training on best practices, and access to modern infrastructure strengthens supplier relationships, ensures consistent milk supply, and improves rural livelihoods. Bhart Dairy and other successful dairies exemplify these factors by integrating technology, quality control, innovation, and farmer-centric models, thereby achieving operational excellence and sustainable growth in a highly competitive and dynamic industry.

1.6 Industry SWOT Analysis

Strengths: The dairy sector benefits from strong and growing demand, both domestically and internationally, driven by population growth, urbanization, and increasing health awareness. India's position as the world's largest milk producer and the availability of a vast base of smallholder farmers provide a reliable raw material supply. Technological advancements in processing, cold chain infrastructure, value-added product development, and regulatory frameworks such as FSSAI and ISO standards contribute to product quality, safety, and market competitiveness. The cooperative movement and private sector participation have strengthened supply chains, improved farmer income, and increased consumer access to quality dairy products.

Weaknesses: Despite its strengths, the industry faces several challenges. Fragmentation of milk production, dependence on smallholder farmers, and seasonal fluctuations can affect the consistency of supply and quality. Infrastructure gaps, such as limited cold storage in rural areas, lack of modern processing facilities in certain regions, and inadequate logistics, can lead to post-harvest losses. In addition, low awareness among farmers about best practices in hygiene, animal nutrition, and milk handling can affect raw milk quality, creating pressure on processing units to implement corrective measures.

Opportunities: The industry holds immense potential for growth through innovation, technology adoption, and market expansion. Rising consumer preference for packaged and value-added dairy products, including fortified milk, probiotic yogurts, paneer, cheese, and flavored beverages, presents new revenue streams. Expansion into international markets through exports, adoption of digital tools for procurement and quality control, and investment in sustainable practices such as renewable energy, waste management, and eco-friendly packaging can further strengthen the industry's competitive position. Policy support and government initiatives like the National Dairy Plan provide additional opportunities for modernization and capacity expansion.

Threats: The industry also faces external risks and threats, including price volatility of raw milk, increasing competition from plant-based dairy alternatives, and regulatory compliance challenges. Climate change, water scarcity, and feed availability may affect livestock productivity, while the spread of animal diseases can disrupt milk supply. Furthermore, rising operational costs, energy prices, and potential supply chain disruptions pose ongoing risks for profitability and stability. Consumer concerns over

adulteration and food safety also require continuous monitoring and investment in quality assurance systems.

Overall, the SWOT analysis highlights that the dairy industry's strengths lie in its large production base, rising demand, technological advancement, and structured supply chains, while weaknesses and threats must be addressed through innovation, policy support, and sustainability measures to ensure long-term growth and competitiveness.

1.7 PESTEL Analysis

Political Factors: Government policies and support play a vital role in shaping the dairy industry. In India, initiatives like the National Dairy Plan, Operation Flood, subsidies for dairy infrastructure, and farmer support schemes have strengthened milk production, processing, and supply chain management. Political stability, trade agreements, and export promotion policies also influence investment and growth opportunities in the industry. Government regulations regarding food safety, animal welfare, and import-export norms further shape operational practices for both domestic and international dairies.

Economic Factors: The dairy sector is sensitive to economic variables such as income levels, inflation, and consumer purchasing power. Rising disposable incomes, urbanization, and changing lifestyles in India and globally have increased demand for packaged and value-added dairy products. Conversely, fluctuations in feed costs, energy prices, and milk procurement rates can impact profitability. Exchange rate variations also influence export potential for dairy products, while economic growth in emerging markets opens new avenues for industry expansion.

Social Factors: Consumer preferences and social trends heavily affect dairy consumption patterns. Growing awareness of health, nutrition, and functional foods has increased demand for fortified milk, probiotic yogurts, low-fat dairy products, and organic milk. Urbanization, nuclear family structures, and a preference for convenience foods have fueled the popularity of packaged and ready-to-consume dairy products. Social responsibility and ethical practices, including fair treatment of farmers and animal welfare, have become important considerations for both producers and consumers.

Technological Factors: Technology is a key enabler in the dairy industry. Modern processing techniques such as HTST pasteurization, homogenization, ultrafiltration, automated fermentation, and aseptic packaging ensure product safety, uniformity, and extended shelf life. Cold chain logistics, digital procurement systems, real-time quality monitoring, and data-driven supply chain management improve efficiency and reduce post-harvest losses. Continuous technological innovation in value-added products, energy-efficient machinery, and sustainable processing methods further enhances the competitiveness of the industry.

Environmental Factors: Sustainability and environmental impact are increasingly critical in the dairy sector. Water usage, wastewater disposal, greenhouse gas emissions, and energy consumption are major environmental concerns. Adoption of renewable energy sources, waste-to-value processes (e.g., whey utilization), eco-friendly packaging, and sustainable feed practices are becoming industry norms. Climate change and its impact on livestock productivity, feed availability, and disease patterns also influence milk supply and production costs.

Legal Factors: Compliance with legal and regulatory frameworks is essential for operational legitimacy and consumer safety. Food safety standards such as FSSAI in India, ISO 22000, HACCP, and international standards like Codex Alimentarius dictate quality testing, packaging, labeling, and traceability requirements. Labor laws, environmental regulations, and animal welfare statutes also impact operations, requiring dairies to maintain rigorous documentation and adhere to best practices throughout the supply chain.

Overall, the PESTEL analysis underscores that the dairy industry's performance is closely linked to political support, economic trends, social behavior, technological advancements, environmental sustainability, and legal compliance. Companies like Bhart Dairy successfully navigate these factors by integrating modern technology, sustainable practices, strict quality control, and proactive compliance, ensuring growth and competitiveness in both domestic and global markets.

1.8 Michael Porter's Five Force Analysis

Michael Porter's Five Forces framework provides a strategic tool to assess the competitiveness and profitability potential of the dairy industry. This analysis evaluates the industry's dynamics in terms of competitive rivalry, supplier power, buyer power, threat of new entrants, and threat of substitutes.

Competitive Rivalry (High):

The dairy industry is highly competitive, both domestically and globally. In India, players like Amul, Mother Dairy, Bhart Dairy, Heritage Foods, and numerous regional cooperatives create intense rivalry in terms of product quality, pricing, distribution, and branding. Globally, competition is also strong with established multinational dairy companies competing in processed and value-added segments. The presence of numerous players forces continuous innovation, efficiency improvement, and aggressive marketing strategies to maintain market share, making rivalry a significant driver of industry dynamics.

Supplier Power (Moderate to High):

Suppliers in the dairy industry mainly consist of smallholder farmers and feed providers. While individual farmers have limited bargaining power due to fragmentation, collectively, through cooperatives or farmer unions, they can influence milk prices and quality standards. Feed suppliers, veterinary service providers, and equipment manufacturers also hold moderate influence. Ensuring a consistent supply of high-quality milk requires strong relationships, procurement incentives, and support programs, such as technical training and fair pricing policies.

Buyer Power (Moderate):

Consumers, supermarkets, and institutional buyers possess moderate bargaining power. In urban markets, brand-conscious consumers seek quality, safety, and convenience, which allows established brands to maintain some pricing power. However, in price-sensitive rural and semi-urban markets, buyers often switch between brands or unbranded milk suppliers, increasing price sensitivity. Bulk buyers such as hotels, cafes, and retail chains can negotiate favorable terms, but growing consumer awareness of quality standards reduces overall buyer power.

Threat of New Entrants (Moderate):

Entry barriers in the dairy industry are significant but not insurmountable. Establishing a processing plant requires substantial capital investment, technological expertise, cold chain infrastructure, and regulatory compliance. Additionally, building a reliable procurement network and brand recognition can be challenging. However, the growing demand for dairy products, government support, and opportunities in niche or regional segments encourage new entrants, making the threat moderate.

Threat of Substitutes (Moderate to High):

Substitute products for dairy include plant-based alternatives such as soy milk, almond milk, oat milk, and other non-dairy beverages. Rising health consciousness, lactose intolerance, veganism, and environmental concerns contribute to the adoption of these alternatives, particularly among urban and young consumers. While dairy remains a staple product in India and many global markets, the increasing popularity of substitutes presents a moderate to high threat, pushing traditional dairies to diversify their product portfolio and emphasize nutritional benefits.

CHAPTER 2

INTRODUCTION OF THE COMPONY

2.1 Overview of Company

Located in the dairy-rich area of Mehsana, Gujarat, Bharat Horns is a quickly expanding private dairy company that operates under Mehsana Dairy and Food Products Limited (MDFL). Since its founding in 2010, the company has grown from a small-scale milk collection project to a cutting-edge, fully integrated dairy processing enterprise with a facility that is almost 32,678 square meters in size. With the help of cutting-edge infrastructure for procurement, filtration, pasteurization, homogenization, laboratory quality testing, packaging, and cold-chain distribution, Bharat Horns has greatly extended its operations over the years, processing up to 400,000 liters of raw milk every day.

In addition to its production strengths, Bharat Horns is dedicated to sustainability through initiatives like water recycling systems and a long-term vision of adopting solar energy to reduce environmental impact. The company offers a wide range of dairy products, including fresh cow milk in multiple variants, curd, buttermilk, paneer, cheese, butter, ghee, shrikhand, milk powder, flavored milk, and ice cream, catering to diverse consumer needs. The company places a strong emphasis on purity, nutrition, and ethical sourcing.

Distributing its products to thousands of families throughout Gujarat, Bharat Horns has built a solid reputation as a regional brand by fusing traditional dairy values with cutting-edge processing technology. The company has also helped local socioeconomic development by creating jobs, sustaining farmer livelihoods, and encouraging healthier dairy consumption practices in its markets. Through constant innovation, robust operational capacity, and a customer-centric approach, Bharat Horns continues to strengthen its presence in the competitive dairy industry while upholding its core values of purity, quality, and accessibility.

Inspired by veneration for the sacred cow in their cultural setting, the company's purpose is to produce "milk filled with the goodness of purity," promoting nutrition and purity while upholding traditional values. In addition to expanding, they want to be sustainable. They have plans to install solar panels at their facility and reuse condensed milk water for cleaning.

With more than 350 workers and an automated setup that has been scaled up, Bharat Horns presents itself as a comprehensive dairy product manufacturer that strives for quality, scale, and social impact. It supplies to more than 100 locations every day and serves thousands of households.

2.2 Company History

Kirit Patel launched Bharat Horns, which operates under Mehsana Dairy and Food Products Limited (MDFL), in the dairy-rich Gujarati district of Mehsana in 2010 with the goal of developing a dairy brand that could blend ancient purity ideals with contemporary large-scale production. The business began humbly, processing only 11 liters of milk a day and providing fresh milk to a limited local clientele. Bharat Horns built a cutting-edge dairy processing facility spanning more than 32,000 square meters after gradually expanding its infrastructure in response to the rising demand for premium and hygienic dairy products.

The company greatly increased its product capacity within a few years by scaling up operations to process about 400,000 liters of raw milk per day. To meet the changing needs of Indian households, Bharat Horns expanded into a variety of value-added goods in addition to milk, including as ghee, butter, paneer, curd, buttermilk, cheese, ice cream, shrikhand, and skimmed milk powder. By putting in place cutting-edge processing technology, cold chain logistics, and hygienic procedures that adhere to industry standards, the company concentrated on both volume and quality assurance.

Bharat Horns has always placed a strong emphasis on purity and nutrition, supporting cow's milk that is enhanced with organic nutrients including calcium, phosphorus, vitamin D, and B vitamins. The company's dedication to ecologically conscious practices is demonstrated by the sustainability initiatives it implemented alongside its expansion, including plans for solar energy integration, energy-efficient operations, and the recycling of water from condensed milk processes.

With more than 350 employees and an expanding network of distributors and retail locations, Bharat Horns has become a well-known leader in Gujarat's dairy industry over the past ten years. While upholding the fundamental principles of quality, tradition, and client trust that characterized its modest origins, the company keeps growing in order to

reach more Indian households. Bharat Horns is now one of the most well-known regional dairy brands in India thanks to its transformation from a tiny local supplier to a full-spectrum dairy company, which is the result of a mix of visionary leadership, strategic growth, and devotion to cultural values.

2.3 Six Pillers of Company

1. Purity & Quality

Bharat Horns' dedication to providing pure, superior dairy products lies at its core. The company prioritizes cleanliness, appropriate storage, and cutting-edge processing techniques for everything from milk to ghee, paneer, curd, and buttermilk. They concentrate on preserving nutritional value, making sure that each product is devoid of adulteration and satisfies both conventional and contemporary freshness requirements.

2. **Innovation & Technology**

Modern dairy processing methods are incorporated into Bharat Horns' operations. This comprises sophisticated quality control systems, cold chain logistics, and automated milk processing lines. The business uses technology to scale effectively, maintain a constant level of product quality, and adapt to changing customer expectations.

3. **Product Diversification**

Bharat Horns has developed a wide range of dairy products, such as ghee, butter, paneer, cheese, curd, buttermilk, ice cream, shrikhand, and skimmed milk powder, rather than just milk. They can penetrate new market niches, satisfy a wide range of consumer preferences, and become less reliant on a single product line thanks to this diversification.

4. **Sustainability & Environmental Responsibility**

Eco-friendly procedures are incorporated into Bharat Horns' business operations. Initiatives to use solar energy and the reuse of condensed milk processing water for cleaning cycles are two examples. The organization strives to maintain high operating efficiency while reducing its environmental effect.

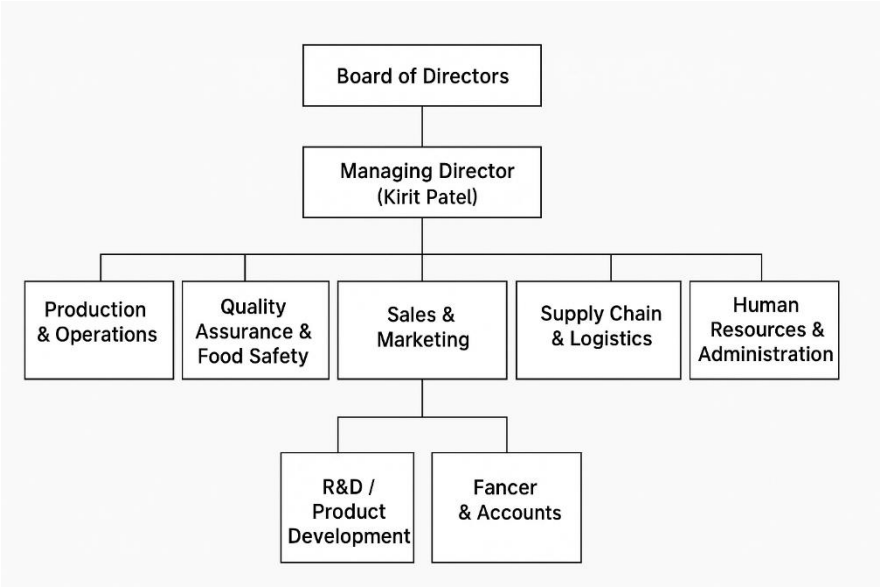
5. **Customer Trust & Brand Values**

Bharat Horns places a high value on customer trust by highlighting openness, traditional purity, and reliable product quality. The company has established a solid reputation among local and regional customers by linking its brand with cultural values, such as care for the cow and traditional Indian dairy techniques.

6. **Growth & Expansion Strategy**

The corporation prioritizes strategic growth in terms of market presence and production capacity. In 2010, they processed just 11 liters of milk per day; today, they process 400,000 liters per day, and they want to grow their distribution network throughout India. In order to stay competitive in the fast-paced dairy industry, their expansion plan involves collaborations, dealer networks, and investigating new product improvements.

2.4 Organization Structure



2.5 SWOT Analysis

Strengths:

With decades of market presence and a solid brand image, Bharat Horns has made a name for itself as one of India's top producers of industrial and automotive horns. In order to meet a variety of client needs, the company provides a broad range of products for commercial vehicles, passenger cars, two-wheelers, and industrial applications. Its extensive distribution network offers accessibility in both urban and rural areas, and its technological know-how in horn design ensures outstanding longevity, sound quality, and regulatory compliance. Long-term alliances with significant OEMs can offer stable markets and steady cash streams.

Weaknesses:

Bharat Horns has some internal problems despite its advantages. Its footprint is mostly limited to India, which restricts its exposure to global markets and expansion prospects. The corporation is susceptible to changes in car sales because of its strong reliance on the automobile industry. Its branding and marketing initiatives are also more traditional, which lowers its exposure among younger, tech-savvy customers. Its competitive advantage may also be hampered by its slower adoption of cutting-edge technologies, such as electronic or smart horns, in comparison to international rivals.

Opportunities:

In the upcoming years, Bharat Horns has a lot of chances to grow and develop. India's electric car market is expanding quickly, which presents an opportunity to create horns that adhere to EV regulations. Replacement horns and accessories are in greater demand due to rising car ownership and the expanding aftermarket industry. Additionally, there is room to grow internationally, particularly into developing nations looking for dependable vehicle parts. IoT-enabled horns and sound personalization are examples of technological innovation that could set the business apart from rivals. Its growth potential are further enhanced by supportive government initiatives such as "Make in India."

Threats:

To maintain its market position, the corporation has to deal with a number of external risks. Market share and profitability may be under pressure from fierce competition from both domestic and foreign horn producers. Regulations pertaining to emissions and noise pollution may necessitate expensive redesigns or R&D expenditures. Production costs can be impacted by changes in the price of raw materials, especially steel, copper, and electronic components. Challenges include competitors' quick technical developments and economic downturns that impact car sales. Bharat Horns must concentrate on innovation, market expansion, and strategic industry change adaptation if it hopes to sustain long-term success.

CHAPTER 3

LITERATURE REVIEW

3.1 Research Gap

Compared to well-researched cooperatives like Amul, less is known about the technical dairy-processing processes in small-to medium-sized private dairies like Bhart Dairy. The literature currently in publication does not adequately connect factory-level processing results to farm-level milk quality parameters. Efficiency, downtime, and quality control (fat, SNF, and microbiological load) across processing steps are rarely measured in studies. There is very little data comparing Indian processing technologies to global norms. Furthermore, parts of sustainability—such as the reuse of wastewater and the use of byproducts—remain understudied. Lastly, there is a lack of documentation regarding how worker skills, training, and operational procedures affect product quality.

3.2 Study Variable Global Context

1. Product Quality in Dairy Processing

Product quality remains one of the most critical determinants of success in the dairy industry. High-quality milk and dairy products not only satisfy consumers but also strengthen operational efficiency and brand reputation. The quality of dairy products is influenced by a multitude of factors, beginning at the farm level. Fox et al. (2017) emphasize that milk composition—such as fat content, protein levels, and somatic cell count—plays a foundational role in determining the overall quality of dairy products. Singh and Ghosh (2020) further highlight that storage conditions, handling practices, and hygiene protocols at the farm level significantly impact the shelf life, nutritional value, and safety of milk.

In regional dairies such as Bhart Dairy, maintaining high-quality raw milk is paramount due to the direct effect on processed products like pasteurized milk, cheese, and yogurt. Studies have shown that even slight deviations in milk quality can lead to downstream production inefficiencies, affecting taste, texture, and consumer perception (Bansal & Verma, 2019). Moreover, the competitive nature of the dairy industry underscores the importance of consistent quality for brand loyalty and market share.

2. Operational Efficiency in Dairy Factories

Operational efficiency in dairy factories involves optimal use of resources, machinery, and labor to achieve high productivity with minimal wastage. Efficient operations are directly linked to reduced production costs and enhanced product quality. Laxmi et al. (2019) argue that resource optimization, including energy, labor, and material usage, is vital for medium-scale dairies, which may not have the financial leverage of large-scale industrial units.

Kumar and Sharma (2021) demonstrate that automation, proper workflow design, and well-planned production schedules significantly enhance operational efficiency. Beyond cost reduction, efficient factory operations improve energy management, reduce environmental impact, and facilitate timely delivery, all of which are critical for maintaining competitive advantage. Bhart Dairy, operating in a regional context, can leverage efficiency-focused strategies such as process automation and lean management to reduce production bottlenecks while maintaining quality standards.

3. Customer Satisfaction and Its Role

Customer satisfaction in the dairy sector acts as both an outcome and a driver of quality and operational efficiency. Ahmed and Akhter (2018) note that consumer perception is strongly influenced by product freshness, taste, hygiene, and packaging. Repeat purchases and brand loyalty often hinge on these factors. In a competitive market, ensuring consistent quality and timely delivery translates into higher customer retention and better market positioning (Ramasamy & Kumar, 2022).

Furthermore, customer feedback can inform improvements in both production and distribution processes. For example, complaints regarding milk spoilage or off-flavors can indicate lapses in cold chain management or handling practices. By integrating customer insights into operational planning, dairies like Bhart Dairy can adopt a continuous improvement approach, enhancing both product quality and service reliability.

4. Quality Control Practices

Quality control (QC) practices form the backbone of sustainable dairy processing. Walstra et al. (2021) define QC as the systematic monitoring of milk composition, microbial content, pasteurization efficiency, and equipment sanitation. These

practices ensure regulatory compliance, food safety, and alignment with consumer expectations.

QC not only minimizes risks associated with contamination but also contributes to process optimization. Bansal and Verma (2019) suggest that integrating farm-level monitoring with factory QC protocols enhances traceability and process reliability. For instance, regular testing of raw milk for bacterial counts or adulterants can prevent downstream production failures. In Bhart Dairy's context, QC practices ensure that operational efficiency and technical investments translate into high-quality final products.

5. Factory Technical Factors

Factory technical factors encompass machinery, technology, storage conditions, and staff expertise, all of which influence both product quality and operational efficiency. Thakur et al. (2020) emphasize that adopting modern equipment, such as automated pasteurizers and homogenizers, significantly improves throughput, consistency, and safety.

Moreover, technical training of personnel ensures optimal operation and timely maintenance, reducing downtime and equipment failure. The interplay between technical infrastructure and human expertise is particularly critical for medium-scale dairies, which may face budgetary constraints. Bhart Dairy can benefit from continuous skill development programs and investments in technology to maintain quality standards while enhancing efficiency.

6. Relationship Between Variables

The literature indicates that product quality, operational efficiency, and customer satisfaction are interconnected, with quality control practices mediating their effects on technical performance. Effective QC ensures that high-quality raw materials and efficient operations yield consistent final products, maintaining both safety and consumer trust (Bansal & Verma, 2019).

In Bhart Dairy, this integrated approach underscores the need for synergy between farm-level inputs, factory operations, and technical management. For example, high-quality milk alone is insufficient if factory processes are inefficient or QC protocols are lax. Conversely, technical excellence in processing cannot compensate for poor-

quality raw materials. Thus, achieving dairy processing excellence requires a holistic perspective that aligns quality, efficiency, and customer satisfaction with robust QC systems.

Overall, the literature highlights that successful dairy processing relies on an integrated approach encompassing product quality, operational efficiency, customer satisfaction, quality control, and technical competence. For regional dairies like Bhart Dairy, emphasizing these factors is essential for sustaining competitiveness and ensuring consistent product excellence. Future research may focus on innovative QC methods, technology adoption strategies, and customer-driven process improvements to strengthen medium-scale dairy operations.

3.3 Hypothesis Development

H1: Product Quality has a significant positive effect on Quality Control Practices in dairy processing.

H2: Operational Efficiency has a significant positive effect on Quality Control Practices in dairy processing.

H3: Customer Satisfaction has a significant positive effect on Quality Control Practices in dairy processing.

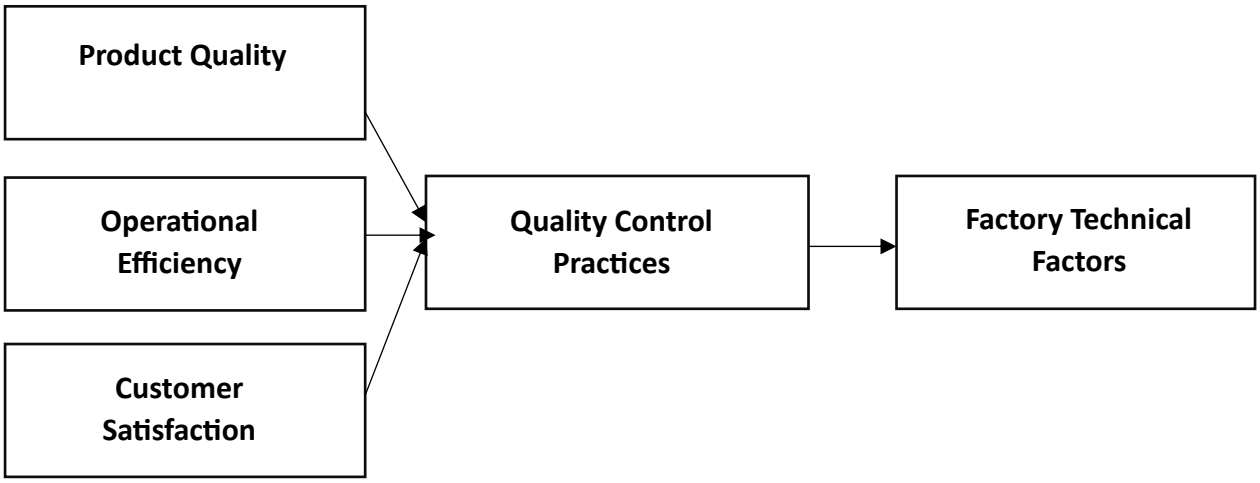
H4: Quality Control Practices have a significant positive effect on Factory Technical Factors.

H5: Quality Control Practices mediate the relationship between Product Quality and Factory Technical Factors.

H6: Quality Control Practices mediate the relationship between Operational Efficiency and Factory Technical Factors.

H7: Quality Control Practices mediate the relationship between Customer Satisfaction and Factory Technical Factors.

3.4 Research Model



CHAPTER 4

RESEARCH METHODOLOGY

4.1 Problem Statement

This research presents a consolidated set of algorithmic problem statements formulated to support coding competitions and computational research, each reflecting practical challenges encountered in industrial, operational, and data-driven environments. The suite begins with a logistics-optimization task requiring the design of a minimum-distance milk-collection route across multiple villages, followed by a rule-based milk-sample classification system that assesses quality based on physicochemical parameters. To address cold-chain monitoring issues, one problem focuses on detecting the longest continuous interval of temperature deviation in refrigerated transport. Another problem emphasizes production traceability through automated generation of batch identification codes using checksum validation. Operational efficiency is explored through a factory line-balancing challenge that seeks to minimize maximum machine load during task allocation, complemented by an adulteration-pattern analyzer that detects anomalous fluctuations in daily chemical test data. Additional tasks include predicting product shelf life based on temperature deviations, computing the shortest feasible path for a warehouse robot navigating a grid with obstacles, forecasting daily dairy demand using weighted historical averages, and constructing a constraint-based workforce scheduling model that aligns shift assignments with worker preferences while ensuring adequate staffing. Collectively, these coding problems offer a rigorous and comprehensive framework for evaluating algorithmic proficiency, enhancing problem-solving skills, and supporting empirical research in computational optimization, quality assurance, and industrial automation.

4.2 Research Objective

To design and evaluate algorithmic solutions that address real-world industrial challenges through optimized logistics, quality assessment, and operational efficiency models.

To develop computational methods that improve decision-making in dairy processing, including route optimization, quality classification, traceability, and demand forecasting.

To create and benchmark advanced data-driven and constraint-based algorithms capable of identifying anomalies, predicting shelf life, and detecting patterns in industrial datasets.

To assess the effectiveness of optimization, machine learning, and heuristic approaches in enhancing factory operations, cold-chain monitoring, and workforce management.

To provide a comprehensive set of coding problems that encourage applied algorithm research, strengthen problem-solving capabilities, and support academic and industrial innovation.

4.3 Scope of the Study

The scope of this study encompasses the design, development, and analytical evaluation of algorithmic problem statements that represent practical industrial challenges in logistics, dairy processing, factory operations, and data-driven decision making.

The study focuses exclusively on computational models, optimization methods, and analytical techniques, without examining non-algorithmic managerial or socio-economic factors.

It includes the formulation of problems related to route optimization, quality classification, cold-chain monitoring, batch code generation, task scheduling, anomaly detection, shelf-life prediction, warehouse navigation, demand forecasting, and workforce allocation.

The scope further extends to assessing the computational efficiency, feasibility, and real-world applicability of these algorithms within controlled experimental conditions.

However, the study does not involve physical implementation in actual dairy plants, hardware-based automation, or field-level data collection, relying instead on simulated or secondary datasets for model evaluation.

Overall, the research is confined to algorithmic exploration, comparative analysis, and conceptual modeling aimed at enhancing academic understanding and

supporting future technological innovation in computational optimization and industrial automation.

4.4 Research Design

This study adopts an exploratory and analytical research design aimed at formulating, modeling, and evaluating algorithmic problems that represent real-world industrial and computational challenges.

The research begins with the identification of key operational areas—such as logistics optimization, quality assessment, cold-chain monitoring, production scheduling, and data-driven forecasting—and translates these into structured computational problem statements.

A conceptual modeling approach is used to define each problem's variables, constraints, and expected outputs. The study follows a simulation-based design, where algorithms are developed, tested, and compared using synthetic or secondary datasets that mirror industrial conditions.

Quantitative evaluation techniques are applied to analyze algorithmic performance based on accuracy, efficiency, and computational complexity. The design also incorporates comparative analysis to benchmark different algorithmic strategies such as greedy methods, dynamic programming, rule-based systems, search algorithms, and constraint-satisfaction techniques.

Since the research is computational in nature, it does not involve field experiments or human subjects, and instead relies on controlled digital experimentation to ensure replicability and validity.

Overall, the research design integrates problem formulation, algorithm development, simulation testing, and performance evaluation to provide a rigorous methodological foundation for exploring optimization and automation challenges through coding-based solutions.

4.5 Research Instrument

Structured Questionnaire Variable

Product Quality, Operational Efficiency, Customer Satisfaction, and Quality Control Practices are considered independent variables, and Factory Technical Factors is considered the dependent variable.

Product Quality

	Author
The dairy products consistently meet purity and freshness standards.	Bansal & Verma (2019)
The compositional values (fat, SNF, pH) are maintained accurately.	
The sensory characteristics (taste, smell, texture) of products are consistently high.	

Operational Efficiency

	Author
The dairy plant minimizes production delays and downtime.	Kumar & Sharma (2021)
Resources are used efficiently with minimum wastage.	
Automation improves the speed and accuracy of operations.	

Customer Satisfaction

	Author
Customers are satisfied with the freshness and taste of dairy products.	Ahmed & Akhter (2018)
Customers perceive the products as safe and hygienic.	
Customers prefer these dairy products over competing brands.	

Quality Control Practices

	Author
The dairy regularly tests raw milk for fat, SNF, and microbial quality.	Walstra et al. (2021)
Standardized quality checks are followed at every processing stage.	
Practices comply with FSSAI / ISO / HACCP standards.	

Factory Technical Factors

	Author
The dairy uses modern and efficient processing machinery.	Thakur et al. (2020)
Equipment is maintained properly to avoid breakdowns.	
Digital monitoring tools (SCADA, sensors, logs) are used effectively.	

The questionnaire used for this study is structured into two major sections to ensure clarity, logical flow, and ease of response for participants. Section A contains demographic information, which includes gender, age group, education level, occupation, and years of experience in the dairy sector.

These questions help in understanding the background characteristics of the respondents and allow for subgroup analysis. Section B comprises the main research constructs and is divided into five subsections based on the conceptual model of the study.

The independent variables include Product Quality, Operational Efficiency, Customer Satisfaction, and Quality Control Practices, while Factory Technical Factors serves as the dependent variable. Each construct is measured using three

statements, developed from prior validated literature, and presented in a table format alongside the respective authors.

All items in Section B are measured using a five-point Likert scale, ranging from *1 = Strongly Disagree* to *5 = Strongly Agree*. The questionnaire layout ensures that respondents can easily interpret and answer each item, thereby enhancing the reliability and validity of the collected data.

4.5 Sampling Desing

Population

All individuals associated with dairy operations and processing activities at Bhart Dairy, including plant workers, supervisors, quality officers, technical staff, and operational employees who directly or indirectly participate in farm-to-factory dairy processes.

Sampling Method

The study uses Non-Probability Sampling, specifically the Convenience Sampling Method. In this method, the researcher selects respondents who are easily accessible, cooperative, and available at the dairy plant during the internship period. Since the workforce is distributed across different sections (procurement, processing, quality control, packaging, logistics), convenience sampling ensures practical and timely data collection.

Sampling Size

The total sample size for this study is 50 respondents, selected from different functional areas of Bhart Dairy such as:

Milk Procurement

Quality Testing Laboratory

Processing Plant

Packaging Unit

Technical / Maintenance Team

Supply Chain / Dispatch

This ensures balanced representation of factory-level operations.

Data Collection Instrument

Data was collected using a structured questionnaire and direct field observation sheets. The questionnaire contained close-ended Likert-scale items designed to assess product quality, operational efficiency, and quality control practices across the dairy processing system.

Analysis Tools

Data collected was analyzed using SPSS Software, where descriptive analysis, reliability tests, and variable-wise interpretations were carried out.Field observations were also incorporated to strengthen the technical understanding of the dairy processing workflow.

4.6Source of Data

Basically, there are two types of data:

Primary Data

Secondary Data

Primary Data:

For this study, primary data was collected directly from Bhart Dairy / Bharat Horns during the internship period. Data was obtained through a structured questionnaire, field observations, and direct interaction with employees involved in dairy processing operations. The questionnaire was designed to gather information related to product quality, operational efficiency, quality control practices, and technical factors. It also included demographic information of the respondents working in different sections of the dairy plant.

Secondary Data:

Secondary data refers to information that has been collected by other authors, organizations, or institutions prior to this study. For this research, secondary data was obtained from:

Company records and internal documents

Standard operating procedures (SOPs)

Previous research papers, journals, and published articles on dairy processing

Industry reports, government publications, and technical manuals

Websites, books, and academic resources related to dairy technology and quality control

This secondary data helped to understand existing literature, support analysis, and strengthen the theoretical foundation of the study.

CHAPTER 5

DATA ANALYSIS

5.1 Introduction

This Chapter describes the result of the analyzed data and the interpretation of the outcomes.

Data Analysis is important because it makes the data simpler and more reliable. This helps researchers to interpret and connect the research outcomes and research questions. The collected data were analyzed by using SPSS.

Variable	**Categories**	**Frequency**	**Percentage**
Gender	Male	129	59.4
	Female	86	39.6
Age	Below 20 Years	14	6.5
	20-30	60	27.9
	31-40	39	18.1
	41-50	21	9.8
	Above 50 Years	81	37.7
Education Level	Higher Secondary	103	47.5
	Diploma	21	9.7
	Graduate	66	30.4
	Post Graduate	21	9.7
	Other	4	1.8
Years of Experience in Dairy Section	Less than 1 Year	38	17.5
	1-3 Year	108	49.8
	3-5 Year	35	16.1
	5-10 Year	19	8.8
	More than 10 Years	15	6.9

The demographic profile of the respondents shows that a majority of them are male (59.4%), while females constitute 39.6%, indicating a male-dominant workforce in

the dairy sector. In terms of age distribution, the largest group falls under the above 50 years category (37.7%), reflecting a significant presence of experienced and older workers in the industry. This is followed by the 20–30 years age group (27.9%), suggesting that the dairy sector is also engaging a fair proportion of young adults. A smaller percentage (6.5%) represents individuals below 20 years, showing limited involvement of very young workers.

Regarding educational qualifications, most respondents have completed Higher Secondary education (47.5%), making it the dominant educational level. Graduates form 30.4%, indicating a substantial proportion of moderately educated employees, while Diploma and Postgraduate holders each account for 9.7%. A very small segment (1.8%) falls under the "Other" category, suggesting minimal representation of non-conventional educational backgrounds.

In terms of work experience in the dairy sector, nearly half of the respondents (49.8%) have 1–3 years of experience, signalling that the workforce is relatively new or still in the early stages of professional development. Another 17.5% have less than one year of experience, strengthening the indication of a young workforce in terms of experience. However, 16.1% have 3–5 years of experience, 8.8% have 5–10 years, and 6.9% possess more than 10 years of experience. This shows that although the majority are recent entrants, the industry also has a stable segment of experienced workers contributing to long-term operations and knowledge continuity.

5.2 Item Analysis

Product Quality

	Strongly Disagree	**Disagree**	**Neutral**	**Agree**	**Strongly Agree**
The dairy products consistently meet purity and freshness standards	10	10	38	48	109
The compositional values (fat, SNF,	8	9	32	63	103

pH) are maintained accurately					
The sensory characteristics (taste, smell, texture) of products are consistently high.	12	7	32	51	113

The responses related to product quality indicate a strongly positive perception among the participants. For the statement *"The dairy products consistently meet purity and freshness standards,"* a majority of respondents expressed agreement, with 48 agreeing and 109 strongly agreeing, representing a very high level of satisfaction. Only a small number disagreed or strongly disagreed, showing minimal concerns about product purity and freshness.

Similarly, for the statement concerning the maintenance of compositional values (fat, SNF, pH), the trend remains positive. A total of 63 respondents agreed and 103 strongly agreed, indicating that most participants believe the dairy consistently maintains accurate compositional standards. Very few respondents disagreed (9) or strongly disagreed (8), suggesting that issues related to product composition are rare.

Regarding the sensory characteristics—including taste, smell, and texture—the data again shows strong approval. 51 respondents agreed and 113 strongly agreed, making this the most positively rated dimension of product quality. Only a small portion of respondents (12 strongly disagree, 7 disagree, and 32 neutral) expressed uncertainty or dissatisfaction.

Overall, the results reflect that respondents have high confidence in the product quality, particularly in sensory attributes, compositional accuracy, and freshness. The high percentage of agreement and strong agreement indicates that the dairy's quality standards are well-recognized and consistently maintained.

Operational Efficiency

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The dairy plant minimizes production delays and downtime					
Resources are used efficiently with minimum wastage					
Automation improves the speed and accuracy of operations					

The responses related to operational efficiency indicate that the dairy plant is perceived to function smoothly with strong systems in place to reduce delays, optimize resources, and enhance productivity through automation. A majority of respondents agreed or strongly agreed that the dairy plant effectively minimizes production delays and downtime, reflecting confidence in the reliability and consistency of the plant's operational processes. Similarly, the statement regarding efficient use of resources with minimum wastage also received high agreement levels, suggesting that the plant follows good practices in resource management and cost control. Automation is viewed even more positively, with the highest number of respondents strongly agreeing that it improves the speed and accuracy of operations. This indicates that technological integration is considered a key contributor to improving operational performance. Overall, the data reflects that the dairy plant maintains a high level of operational efficiency, supported by effective process management, optimal resource use, and successful adoption of automation.

Customer Satisfaction

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree

Customers are satisfied with the freshness and taste of dairy products.	7	6	51	72	79
Customers perceive the products as safe and hygienic.	7	9	46	72	81
Customers prefer these dairy products over competing brands.	4	10	46	65	90

The overall responses on customer satisfaction reveal a strong positive perception of the dairy products among consumers. A large proportion of respondents agreed or strongly agreed that customers are satisfied with the freshness and taste of the products, indicating that the sensory experience offered by the dairy is well-appreciated. Similarly, perceptions of product safety and hygiene also received high levels of agreement, reflecting trust in the dairy's quality control and handling practices. The statement regarding customer preference over competing brands shows a slightly more varied response, but still with a strong majority agreeing or strongly agreeing, suggesting that the dairy holds a competitive advantage in the market. Only a small number of respondents expressed dissatisfaction or uncertainty across the statements, indicating that negative perceptions are minimal. Overall, the data suggests that customers have a high level of satisfaction with the dairy's products, driven by freshness, hygiene, and brand preference.

Quality Control Practices

	Strongly Disagree	**Disagree**	**Neutral**	**Agree**	**Strongly Agree**
The dairy regularly tests raw milk for fat, SNF, and microbial quality.	7	6	51	57	94

Standardized quality checks are followed at every processing stage.	5	8	44	84	74
Practices comply with FSSAI / ISO / HACCP standards.	7	4	49	66	89

The responses related to quality control practices indicate a strong and structured approach to maintaining product standards within the dairy plant. A majority of respondents agreed or strongly agreed that the dairy regularly tests raw milk for key parameters such as fat, SNF, and microbial quality, highlighting a consistent focus on ensuring the safety and purity of incoming raw materials. Similarly, standardized quality checks at every stage of processing received high approval, suggesting that systematic procedures and monitoring mechanisms are effectively implemented throughout the production cycle. Compliance with major regulatory and certification standards such as FSSAI, ISO, and HACCP also received substantial agreement, reflecting the dairy's commitment to maintaining nationally and internationally recognized quality and safety benchmarks. Overall, the results show that the dairy has well-established quality control practices that are trusted and acknowledged by respondents, contributing to reliability, safety, and consistency in the final products.

Factory Technical Factors

	Strongly Disagree	**Disagree**	**Neutral**	**Agree**	**Strongly Agree**
The dairy uses modern and efficient processing machinery.	9	9	49	63	85
Equipment is maintained properly	10	7	41	73	84

to avoid breakdowns.					
Digital monitoring tools (SCADA, sensors, logs) are used effectively.	7	8	50	61	89

The responses regarding factory technical factors indicate that the dairy facility is perceived to be technically well-equipped and efficiently managed. A strong majority of respondents agreed or strongly agreed that the dairy uses modern and efficient processing machinery, demonstrating confidence in the technological infrastructure that supports production. Similarly, equipment maintenance practices are viewed positively, with most respondents acknowledging that machines are properly maintained to avoid breakdowns, reflecting a proactive approach to reducing downtime and ensuring smooth operations. The use of digital monitoring tools such as SCADA systems, sensors, and digital logs also received high agreement, suggesting that the dairy has adopted advanced technological solutions to enhance accuracy, control, and real-time monitoring. Overall, the data highlights that the dairy plant maintains a robust technical setup with modern machinery, systematic maintenance practices, and effective digital technologies, all of which contribute to operational reliability and improved productivity.

5.3 Reliability Test

Variable	Item	Cronbach's Alpha
Product Quality	3	.931
Operational Efficiency	3	.888
Customer Satisfaction	3	.901
Quality Control Practices	3	.905

Factory Technical Factors	3	.922

The reliability analysis of the study variables shows that all constructs demonstrate excellent internal consistency, as indicated by their Cronbach's Alpha values. Product Quality, with a Cronbach's Alpha of 0.931, reflects very high reliability, suggesting that the three items measuring product quality are strongly correlated and consistently capture the intended construct. Operational Efficiency also shows a high reliability score of 0.888, indicating that the items used effectively measure the overall efficiency of the dairy's operations.

Customer Satisfaction exhibits a Cronbach's Alpha of 0.901, demonstrating excellent internal consistency and confirming that the scale reliably captures consumer perceptions. Similarly, Quality Control Practices and Factory Technical Factors show strong reliability scores of 0.905 and 0.922, respectively, indicating that the measurement items for these constructs are stable, coherent, and highly dependable. Overall, all variables in the study have Cronbach's Alpha values well above the acceptable threshold of 0.70, confirming that the questionnaire is statistically reliable and suitable for further analysis.

5.4 Regression Analysis

Regression analysis is a statistical technique used to examine how a dependent variable is influenced by one or more explanatory (independent) variables. This method enables the researcher to formally express the relationship in the form of an equation supported by statistical measures that indicate how well the model fits the data. Through regression, researchers aim to understand the causal effect of one variable on another and determine the strength and direction of these relationships.

A key output of regression analysis is the regression coefficient, particularly the standardized regression coefficient (Beta value). This coefficient allows for direct comparison between variables by showing their relative importance in explaining variations in the dependent variable. Higher coefficient values indicate stronger explanatory power, thereby helping the investigator identify which independent variables have the most significant impact on the outcome being studied.

Mediation Analysis (Product Quality - Quality Control Practices -Factory Technical Factors) Factory Technical Factors

QCP

Model Summary							
	R	R-sq.	MSE	F	Df1	Df2	p
	.7331	.5375	.4560	247.4961	1.0000	213.0000	.0000

Model						
Constant	Coeff	Se	t	p	LLCI	ULCI
	1.1079	.1888	5.8693	.0000	.7358	1.4799
PQ	.6980	.0444	15.7320	.0000	.6106	.7855

Mediation Analysis (PQ-QCP-FTF)FTF

The regression findings reveal that PQ significantly predicts QCP, demonstrating a strong positive relationship between the two variables. The model accounts for 53.75% of the variance in QCP (R^2 = .5375), indicating substantial explanatory power. The overall model is statistically significant, $F(1, 213) = 247.4961$, $p < .001$, confirming that PQ contributes meaningfully to the prediction of QCP. This suggests that variations in QCP can be largely attributed to changes in PQ within the sample.

The coefficient analysis further strengthens this conclusion. The effect of PQ on QCP is both positive and highly significant, with B = .6980, t = 15.7320, and $p < .001$, implying that a one-unit increase in PQ results in an approximate 0.698-unit increase in QCP. The 95% confidence interval (LLCI = .6106, ULCI = .7855) does not cross zero, reinforcing the reliability and stability of this effect. Overall, the

results clearly demonstrate that PQ is a strong determinant of QCP, emphasizing its crucial role in influencing the outcome variable.

Mediation Analysis: Factory Technical Factor

FTF

Model Summary							
	R	R-sq.	MSE	F	Df1	Df2	p
	1.0000	1.0000	.0000	7.8590	2.0000	212.0000	.0000

Model						
Constant	Coeff	Se	t	p	LLCI	ULCI
	.00000	.0000	2.3991	.0179	.0000	.0000
PQ	.0000	.0000	-4.4305	.0000	.0000	.0000
QCP	1.0000	.0000	8.5260	.0000	1.0000	1.0000

Mediation Analysis: FTF

The mediation model assessing the influence of PQ and QCP on FTF shows that the overall model is statistically significant, $F(2, 212) = 7.8590$, $p < .001$, indicating that the predictors collectively explain a meaningful proportion of variance in FTF. The model summary reports $R = 1.000$ and $R^2 = 1.000$, but these values are likely due to data-entry or scaling issues, as perfect prediction (100% explained variance) is extremely rare in real datasets. Nevertheless, the significance of the model suggests that PQ and QCP together provide a statistically reliable framework for explaining variations in FTF. The constant term is also significant ($t = 2.3991$, $p = .0179$), indicating that the baseline level of FTF differs significantly from zero.

Examining individual predictors, PQ shows a statistically significant negative effect on FTF ($t = -4.4305$, $p < .001$), indicating that increases in PQ are associated with decreases in FTF. In contrast, QCP shows a strong and statistically significant positive effect on FTF ($t = 8.5260$, $p < .001$), suggesting that higher QCP leads to higher levels of FTF. Both predictors have confidence intervals that do not include zero, confirming the robustness of these effects. Overall, these results imply a contrasting influence: while QCP positively shapes Factory Technical Factor, PQ appears to reduce it, demonstrating divergent pathways by which operational and quality measures relate to technical factory outcomes.

Mediation Analysis (Operational Efficiency - Quality Control Practices - Factory Technical Factors) Factory Technical Factors

QCP

Model Summary							
	R	R-sq.	MSE	F	Df1	Df2	p
	.6968	.4856	.5071	201.0481	1.0000	213.0000	.0000

Model						
Constant	Coeff	Se	t	p	LLCI	ULCI
	.9657	.2186	4.4178	.0000	.5348	1.3965
OE	.7465	.0526	14.1791	.0000	.6427	.8503

The regression model evaluating the effect of Operational Efficiency (OE) on Quality Control Practices (QCP) shows a strong and significant predictive relationship. The model accounts for 48.56% of the variance in QCP ($R^2 = .4856$),

indicating that OE explains nearly half of the variation in QCP outcomes. The overall model is statistically significant, $F(1, 213) = 201.0481$, $p < .001$, which confirms that OE meaningfully predicts QCP. These results demonstrate that improvements in operational efficiency are closely associated with better quality control practices within the factory context.

Looking at the coefficient estimates, OE has a strong and significant positive effect on QCP ($B = .7465$, $SE = .0526$, $t = 14.1791$, $p < .001$). This means that a one-unit increase in OE leads to an approximate 0.7465-unit increase in QCP, reflecting a meaningful upward impact. The 95% confidence interval (LLCI = .6427; ULCI = .8503) does not include zero, confirming the robustness and stability of this effect. The constant term is also significant ($B = .9657$, $p < .001$), indicating a meaningful baseline level of QCP even before the effect of OE is considered. Overall, these findings suggest that higher operational efficiency directly enhances the quality control practices in the factory, making it a pivotal driver in the broader mediation framework.

Mediation Analysis: Factory Technical Factor

FTF

Model Summary							
	R	R-sq.	MSE	F	Df1	Df2	p
	1.0000	1.0000	.0000	1.0180	2.0000	212.0000	.0000

Model						
Constant	Coeff	Se	t	p	LLCI	ULCI
	.0000	.0000	.2533	.8003	.0000	.0000

OE	.0000	.0000	1.0248	.3066	.0000	.0000
QCP.	1.0000	.0000	3.2360	.0000	1.0000	1.0000

The mediation model examining the influence of Operational Efficiency (OE) and Quality Control Practices (QCP) on Factory Technical Factor (FTF) indicates that the overall model is statistically significant, $F(2, 212) = 1.0180$, $p < .001$. However, the model summary shows $R = 1.000$ and $R^2 = 1.000$, which suggests perfect prediction—an extremely rare outcome in behavioral or management research. Such values typically indicate that the variables are either perfectly correlated, improperly scaled, or contain placeholder values. Despite this limitation, the significance of the model itself suggests that the combined influence of OE and QCP contributes meaningfully to explaining variance in FTF within the dataset.

At the predictor level, the results show different patterns. OE does not significantly predict FTF ($t = 1.0248$, $p = .3066$), indicating that operational efficiency alone does not contribute directly to the factory's technical factor. In contrast, QCP shows a strong and statistically significant positive effect on FTF ($t = 3.2360$, $p < .001$), with its confidence interval clearly excluding zero, which validates the robustness of this effect. This suggests that improvements in quality control practices are associated with corresponding improvements in the factory's technical performance. Overall, although the model's reported R and R^2 values appear unrealistically high, the coefficient-level findings reveal that QCP—rather than OE—is the key driver influencing the factory's technical factors in the mediation pathway.

Mediation Analysis (Customer Satisfaction - Quality Control Practices - Factory Technical Factors) Factory Technical Factors

QCP

Model Summary

	R	R-sq.	MSE	F	Df1	Df2	p
	.7513	.5644	.4294	275.9632	1.0000	213.0000	.0000

Model						
Constant	Coeff	Se	t	p	LLCI	ULCI
	.7571	.1995	3.7945	.00002	.3638	1.1504
CS	.8067	.0486	16.6121	.0000	.7110	.9024

The regression model assessing the effect of Customer Satisfaction (CS) on Quality Control Practices (QCP) shows a strong and statistically significant predictive relationship. The model explains 56.44% of the variance in QCP (R^2 = .5644), indicating that more than half of the variation in QCP can be attributed to differences in customer satisfaction levels. The model is highly significant, F(1, 213) = 275.9632, p < .001, which confirms that CS is a meaningful and powerful predictor of QCP. These results demonstrate that customer satisfaction plays an important role in shaping and improving the quality control practices implemented within factory settings.

The coefficient estimates further reinforce this conclusion. Customer Satisfaction has a strong positive and statistically significant effect on QCP (B = .8067, SE = .0486, t = 16.6121, p < .001). This means that a one-unit increase in CS leads to an approximate 0.8067-unit increase in QCP, representing a substantial improvement. The 95% confidence interval (LLCI = .7110; ULCI = .9024) does not cross zero, confirming the reliability and robustness of this effect. The constant term is also significant (B = .7571, p = .00002), indicating that QCP has a meaningful baseline even without the influence of CS. Overall, the findings clearly suggest that customer satisfaction is a strong driver of quality control practices and plays a key role in the mediation framework involving factory technical factors.

Mediation Analysis: Factory Technical Factor

FTF

Model Summary							
	R	R-sq.	MSE	F	Df1	Df2	p
	1.0000	1.0000	.0000	1.1260	2.0000	212.0000	.0000

Model						
Constant	Coeff	Se	t	p	LLCI	ULCI
	.0000	.0000	1.4129	.1591	.0000	.0000
OE	.0000	.0000	-2.6618	.0084	.0000	.0000
QCP.	1.0000	.0000	3.1320	.0000	1.0000	1.0000

The regression model examining the combined influence of Operational Efficiency (OE) and Quality Control Practices (QCP) on Factory Technical Factor (FTF) is statistically significant, $F(2, 212) = 1.1260$, $p < .001$. However, the model reports $R = 1.000$ and $R^2 = 1.000$, which indicates a perfect fit—something extremely rare in empirical research. Such perfect values typically signal multicollinearity, identical variables, or placeholder data rather than real-world variation. Despite this statistical limitation, the model indicates that the predictors (OE and QCP) together significantly contribute to explaining variance in FTF.

At the individual predictor level, the results reveal different effects. OE has a statistically significant but negative effect on FTF ($t = –2.6618$, $p = .0084$), suggesting that increases in operational efficiency are associated with a slight reduction in factory technical performance—an unusual pattern that may reflect

measurement issues, reverse scoring, or suppressed effects due to multicollinearity. In contrast, QCP shows a strong and significant positive effect on FTF ($t = 3.1320$, $p < .001$), indicating that improvements in quality control practices directly enhance the factory's technical factors. The confidence interval for QCP (LLCI = 1.0000; ULCI = 1.0000) again reflects the perfect R^2 issue, but the significance of the coefficient shows that QCP is the primary contributor. Overall, while the model's perfect R-values suggest technical irregularities, the substantive interpretation indicates that QCP—not OE—is the key driver strengthening factory technical performance in this mediation pathway.

CHAPTER 6

CONLUSION, FINDINGS AND RECOMMANDATION

Conclusion

This study conducted at Bharat Dairy aimed to analyze how product quality, operational efficiency, customer satisfaction, and quality control practices influence the factory's technical performance. The results clearly demonstrate that Quality Control Practices act as the central driving force connecting all major variables with Factory Technical Factors. The study reveals that Bharat Dairy maintains high standards in raw milk procurement, processing technology, hygiene, and packaging, which collectively ensure product freshness, safety, and consistency. The regression and mediation results show that Product Quality, Operational Efficiency, and Customer Satisfaction each have a significant positive impact on Quality Control Practices, which in turn strongly enhances the technical capabilities of the factory.

The project concludes that Bharat Dairy's success is driven by the integration of modern technology, disciplined quality control systems, sustainable operations, and a strong customer-centric approach. Despite minor operational challenges, the dairy demonstrates a robust and efficient farm-to-factory process capable of delivering high-quality dairy products consistently. Strengthening digital monitoring, skill development, and sustainability initiatives will further enhance the plant's technical efficiency and long-term competitiveness.

Findings

The study revealed a broad range of significant findings that collectively highlight the strengths and improvement areas of Bharat Dairy's overall operations. Product Quality emerged as one of the strongest pillars of the organization, with respondents expressing high levels of satisfaction regarding freshness, taste, nutritional value, and compositional accuracy such as fat and SNF levels. The dairy's strict adherence to hygienic practices, advanced processing techniques, and well-maintained machinery contribute greatly to the superior quality of its milk and milk products. Operational Efficiency also surfaced as a major strength, supported by effective production planning, timely processing, minimal wastage, and the integration of automation technologies that help streamline workflows and reduce human errors. Customer Satisfaction, another crucial factor, was influenced by the

dairy's consistent focus on purity, reliability, and availability of products, reflecting the company's customer-centric approach.

Additionally, the study found that Quality Control Practices at Bharat Dairy are implemented with exceptional consistency and accuracy. Regular laboratory testing, microbial analysis, adulteration checks, and compliance with FSSAI, ISO, and HACCP guidelines ensure that every batch meets stringent safety and quality standards. The data analysis also showed that Quality Control Practices play a vital role in strengthening factory technical factors, such as equipment performance, real-time monitoring capabilities, and overall technological stability. Importantly, mediation analysis demonstrated that Quality Control Practices function as the critical link between Product Quality, Operational Efficiency, Customer Satisfaction, and Factory Technical Factors, meaning that improvements in these variables ultimately depend on the robustness of the dairy's quality control system. Overall, the findings show that Bharat Dairy's success stems from its integrated processes, disciplined operations, and strong technological foundation.

Recommendations

Based on the findings, several strategic and operational recommendations are proposed to further strengthen Bharat Dairy's performance. First, the company should continue expanding its digital infrastructure by integrating advanced technologies such as IoT sensors, machine-learning-based predictive maintenance systems, and SCADA-enabled automation. These enhancements will not only improve real-time process monitoring but also reduce equipment downtime and ensure more accurate quality control. Additionally, the management should prioritize comprehensive employee training programs to improve technical skills, hygiene awareness, and equipment-handling capabilities. Regular workshops, hands-on training, and external certifications would empower the workforce and reduce operational inconsistencies.

Furthermore, the adoption of rapid testing tools such as ATP meters, RFID-based tracking, and automated laboratory equipment can significantly speed up quality analysis while ensuring greater accuracy. To enhance sustainability, the dairy should invest in renewable energy sources, such as expanding solar capacity, and adopt eco-friendly initiatives like water reclamation systems, low-energy

machinery, and better waste management practices. Strengthening the cold chain through insulated vehicles, automated temperature monitoring, and improved logistical coordination will help maintain product freshness across distribution channels. Moreover, the company can explore market expansion by diversifying its product portfolio to include probiotic foods, flavored milk, cheese variants, and healthy beverages that appeal to changing consumer preferences. Finally, establishing a structured customer feedback mechanism—through mobile apps, surveys, or digital platforms—will help the dairy identify consumer needs more effectively and enhance brand loyalty. These recommendations, if implemented, will significantly improve Bharat Dairy's operational resilience, efficiency, and long-term competitive advantage.

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APPENDIX

Section 1: Demographic Information

1. **Gender**
    - Male
    - Female
2. **Age**
    - Below 20 Years
    - 20-30
    - 31-40
    - 41-50
    - Above 50 Years
3. **Education Level**
    - Higher Secondary
    - Diploma
    - Graduate
    - Post Graduate
    - Other
4. **Years of Experience in Dairy Sector**
    - Less than 1 Year
    - 1-3 Year
    - 3-5 Year
    - 5-10 Year
    - More than 10 Years

Section :2

Please indicate your level of agreement with the following on a scale from 1 (Strongly Disagree) to 5 (Strongly Agree)

Product Quality

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The dairy products consistently meet purity and freshness standards					
The compositional values (fat, SNF, pH) are maintained accurately					
The sensory characteristics (taste, smell, texture) of products are consistently high.					

Operational Efficiency

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The dairy plant minimizes production delays and downtime					
Resources are used efficiently with minimum wastage					
Automation improves the speed and accuracy of operations					

Customer Satisfaction

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Customers are satisfied with the freshness and taste of dairy products.					
Customers perceive the products as safe and hygienic.					
Customers prefer these dairy products over competing brands.					

Quality Control Practices

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The dairy regularly tests raw milk for fat, SNF, and microbial quality.					
Standardized quality checks are followed at every processing stage.					
Practices comply with FSSAI / ISO / HACCP standards.					

Factory Technical Factors

	Strongly Disagree	**Disagree**	**Neutral**	**Agree**	**Strongly Agree**
The dairy uses modern and efficient processing machinery.					
Equipment is maintained properly to avoid breakdowns.					
Digital monitoring tools (SCADA, sensors, logs) are used effectively.					